

UNDERSTANDING SOCIAL ANXIETY

A data-driven investigation into the lifestyle, physiological
and psychological factors that travel with anxiety

PROJECT TEAM

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2,030 RESPONDENTS	22 RAW VARIABLES	18 HYPOTHESIS TESTS	60+ VISUALISATIONS
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Executive Summary

We set out to answer a deceptively simple question: **what separates an anxious person from a calm one?** To answer it we took a survey of 2,030 people covering sleep, diet, exercise, caffeine, alcohol, heart rate, breathing, occupation and therapy history, cleaned it, visualised it from every angle, and then subjected the patterns we found to formal statistical testing.

The short answer is that anxiety in this population is **behavioural, not demographic**. Age does not predict it. Gender does not predict it. What does travel with anxiety is a cluster of habits and physiological signals - above all, stress and sleep.

6.7 hMEAN SLEEP
vs 7 h recommended**294 mg**MEAN CAFFEINE
vs 250 mg baseline**92 bpm**MEAN HEART RATE
vs 72 bpm norm**46.8%**REPORT
HIGH STRESS

The five headline findings

FINDING 1 - STRESS IS THE STRONGEST SIGNAL WE FOUND

Stress and anxiety move together more tightly than any other pair of variables in the dataset (Spearman rho = 0.70, $p < 0.001$). Nothing else comes close. If you want to predict someone's anxiety score from a single number, ask about their stress.

FINDING 2 - SLEEP IS THE clearest protective factor

More sleep goes with less anxiety (rho = -0.34, $p < 0.001$), and the sample is chronically short of it: 63% sleep less than 7 hours a night, and the average of 6.67 hours is significantly below the recommended 7 ($t = -12.43$, $p < 0.001$).

FINDING 3 - THIS IS A PHYSIOLOGICALLY AROUSED SAMPLE

Mean heart rate is 92 bpm against a global norm of 72 ($t = 47.80$, $p < 0.001$), and 54% show a rapid breathing rate. People who experience dizziness report significantly higher anxiety ($p = 0.019$). The body is telling the same story as the questionnaire.

FINDING 4 - DEMOGRAPHY IS A DEAD END

Age has no significant relationship with anxiety (rho = -0.03, $p = 0.154$). Gender is statistically independent of both smoking ($p = 0.622$) and therapy participation ($p = 0.944$). Who you are matters far less than how you live.

FINDING 5 - THE THERAPY RESULT IS A TRAP FOR THE UNWARY

Anxiety level and therapy participation are strongly associated (chi-square = 54.69, $p < 0.001$) - but the arrow almost certainly runs backwards. People seek therapy *because* they are anxious. A cross-sectional survey cannot separate treatment effect from selection effect, and we do not claim it can.

How to read this report

Chapter 1 introduces the data and the question. Chapter 2 documents how the raw file was turned into trustworthy evidence - the least glamorous and most important part of the project. Chapter 3 paints a portrait of the 2,030 people. Chapter 4 follows the threads between variables. Chapter 5 puts every claim through a formal statistical test. Chapter 6 draws the conclusions, and Chapter 7 is an unusually candid account of the limits of what we found, including three data-quality problems we uncovered in our own pipeline.

CHAPTER ONE

The Question and the Evidence

Anxiety is easy to describe and hard to measure. It shows up in how well people sleep, how fast their hearts beat, what they drink, whether they exercise, and whether they have ever sat in a therapist's office - and each of those signals is noisy on its own. The promise of a dataset like this one is that the signals can be read together.

The dataset

Our source is a survey of 2,030 respondents described by 22 variables. They fall into four natural families, and this grouping drove every subsequent decision about how each column should be cleaned, plotted and tested.

Family	Variables	Examples
Numeric	8	Age, Sleep Hours, Caffeine Intake, Heart Rate, Breathing Rate
Ordinal (ranked scales)	4	Stress Level (1-10), Anxiety Level (1-10), Diet Quality (1-10), Sweating Level (1-5)
Categorical	8	Gender, Occupation, Smoking, Medication, Dizziness, Therapy History
Target	1	Target (1 = anxious, 0 = not anxious)

The binary **Target** label is heavily imbalanced: 210 respondents (10.3%) are flagged as anxious against 1,820 (89.7%) who are not. A duplicate label column, *is_Anxious*, was found to be identical to Target and dropped.

That imbalance matters. With only 210 positive cases, any comparison that splits the sample by Target is working with thin slices. For this reason the analytical spine of this report is the continuous **Anxiety Level (1-10)** scale rather than the binary flag: it uses all 2,030 observations and carries far more information.

Missing data: the first surprise

A null audit turned up 2,431 missing values - and they were not evenly spread. Seven columns had modest gaps of 2-6%. One column had almost nothing but gaps.

Column	Missing	% of rows	Treatment
Therapy History	1,827	90.0%	Repaired using therapy-session logic
Gender	119	5.9%	Mode imputation
Medication	109	5.4%	Mode imputation
Smoking	106	5.2%	Mode imputation
Caffeine Intake (mg/day)	94	4.6%	Median (skewed)
Diet Quality (1-10)	78	3.8%	Median (ordinal)
Age	62	3.1%	Mean or median by skew

Column	Missing	% of rows	Treatment
Sleep Hours	36	1.8%	Mean or median by skew
All other columns	0	0.0%	-

WHY 90% MISSING DID NOT MEAN 90% LOST

Therapy History was blank for 1,827 of 2,030 people. Deleting the column would have thrown away the therapy dimension entirely; imputing the most common value would have invented 1,827 therapy histories out of nothing. Neither is acceptable. The gaps turned out not to be random at all - they were recoverable from a neighbouring column, which is the subject of the next chapter.

CHAPTER TWO

From Raw File to Trustworthy Evidence

Every chart and every p-value in this report rests on the decisions described in this chapter. We wrote each step as a reusable function rather than as inline code, so that the entire pipeline is auditable and can be re-applied to new data in a few lines.

Step 1 - Repairing therapy history with domain logic

The missing values in *Therapy History* were not random noise. A neighbouring column, *Therapy Sessions (per month)*, was complete for every respondent - and it carries the same information in a different form. If somebody attends four sessions a month, we know their therapy history is not empty; it simply was not written down. If somebody attends zero sessions, we know why the field is blank.

Condition	Inference	Value assigned
Sessions > 0, history blank	Therapy exists but was not recorded	Not Reported
Sessions = 0	There is genuinely no therapy history	No previous history

This single rule recovered the therapy dimension for the whole sample and kept the two therapy columns internally consistent - something no generic imputation strategy would have achieved.

Step 2 - Removing the impossible

Three numeric columns contained negative values - 120 of them in total. A respondent who sleeps minus ten hours a night is not an unusual case; it is a data-entry error. Each of Sleep Hours, Physical Activity and Alcohol Consumption carried exactly 40 such entries, all bottoming out at -10, which suggests a sentinel value that was never decoded rather than random corruption. These were converted to missing values so that the imputation stage would handle them, instead of silently dragging every average downward.

Step 3 - Taming outliers without losing people

Extreme values distort means, correlations and every downstream test. The obvious fix - deleting the offending rows - is also the worst one: it discards every other answer that respondent gave. Instead we applied the interquartile-range rule and *clipped* values to the fence rather than removing them.

THE IQR FENCE

$IQR = Q3 - Q1$. Lower fence = $Q1 - 1.5 \times IQR$. Upper fence = $Q3 + 1.5 \times IQR$. Any value beyond a fence is pulled back to it. Sample size stays at 2,030; extreme influence shrinks.

The combined effect of steps 2 and 3 is best seen as a table of ranges. Every column still has 2,030 rows; what changed is that the impossible and the extreme have been pulled back to the plausible.

Column	Range before	Range after	Negatives removed
Sleep Hours	-10.0 to 11.0	3.50 to 9.90	40

Column	Range before	Range after	Negatives removed
Physical Activity (hrs/week)	-10.0 to 9.2	0.00 to 8.25	40
Alcohol Consumption (drinks/week)	-10.0 to 19.0	0.00 to 19.00	40
Caffeine Intake (mg/day)	0 to 1,500	0.00 to 710.88	0
Heart Rate (bpm)	60 to 220	60.00 to 153.50	0
Therapy Sessions (per month)	0 to 10	0.00 to 6.00	0
Age	18 to 64	18.00 to 64.00	0

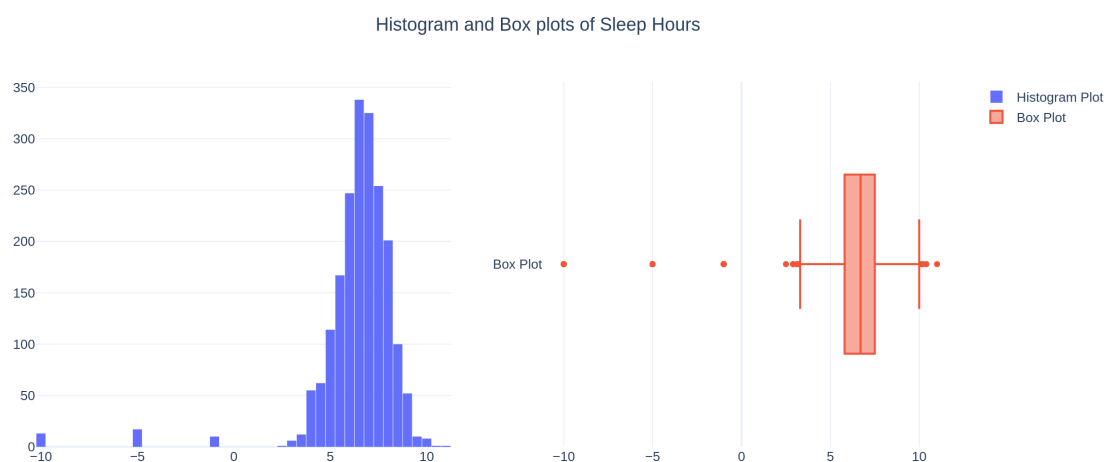


Figure 2.1 - Sleep Hours before cleaning. The distribution runs from -10 to 11 hours; the long left arm of the box plot is made entirely of impossible values.

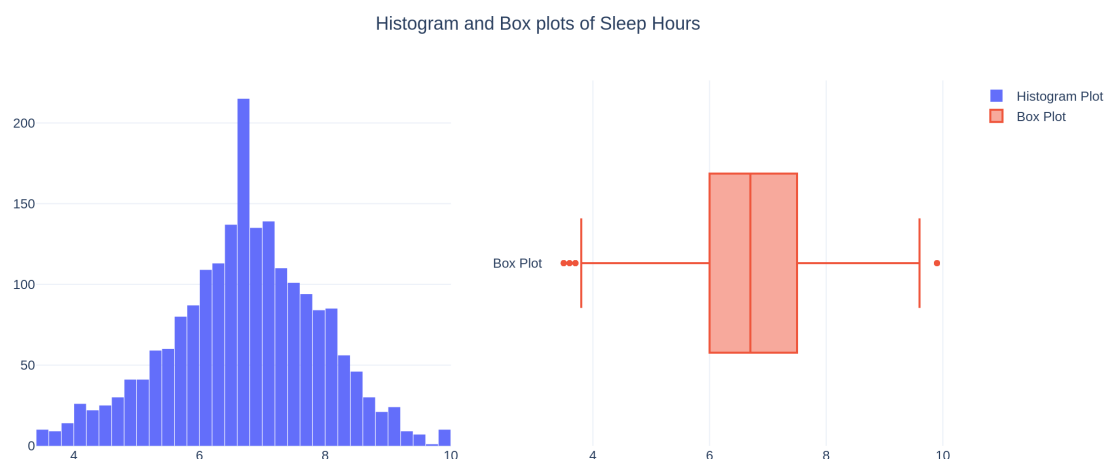


Figure 2.2 - Sleep Hours after cleaning. The same 2,030 respondents, now spanning 3.5 to 9.9 hours and centred on a mean of 6.67.

Step 4 - Imputation that respects the shape of the data

There is no single correct fill value. Using the mean on a heavily skewed column drags every imputed record toward a long tail; using the median on a symmetric column throws away information. So we let each column decide for itself, based on its skewness.

Column type	Condition	Fill value	Reasoning
Numeric	skew < 0.5	Mean	Symmetric - the mean is efficient
Numeric	skew >= 0.5	Median	Skewed - the median resists the tail
Ordinal	always	Median	Returns a real scale point, never a fraction
Categorical	always	Mode	Preserves the existing category balance

MISSINGNESS IS ITSELF INFORMATION

For every imputed column we also created a companion indicator column recording whether the original value was absent. Whether a person declined to answer a question about their diet or their medication may be as informative as the answer itself, and we did not want to erase that signal by filling the gap.

Step 5 - Binning: turning numbers into language

Continuous measurements were also converted into labelled bands using clinically meaningful cut points rather than arbitrary quantiles - sleep splits at 4, 7 and 9 hours; heart rate at 60 and 100 bpm; breathing rate at 12 and 20 breaths per minute. This lets us describe findings in plain language and makes categorical techniques such as chi-square tests and crosstab heatmaps available.

Step 6 - Regrouping and composite indicators

Twelve separate job titles were collapsed into six sectors, so that each group is large enough to support a conclusion. Therapy modalities were collapsed into a simple participation flag. Finally, four composite KPI features were engineered to capture concepts that no single column expresses.

Composite indicator	What it combines	Reads as
Anxiety Risk Index	(Stress x Anxiety) / (Sleep + Activity + Diet)	Psychological load divided by protective habits
Therapy Gap	Anxiety - Therapy Sessions	Unmet need: high anxiety, little treatment
Lifestyle Risk	Count of 5 risk flags	0-5 scale of poor sleep, inactivity, poor diet, heavy drinking, smoking
Mental Health Risk Score	Weighted sum of shortfalls	Symptom load plus distance from each healthy target

The result of all six steps is a clean, fully populated dataset of 2,030 rows, exported once so that every analytical chapter that follows reads from an identical starting point.

CHAPTER THREE

A Portrait of 2,030 People

Before asking what causes what, it is worth simply looking at who these people are. The picture that emerges is remarkably consistent, and slightly alarming: this is a population that is tired, over-caffeinated, physically aroused and under considerable stress.

They are not sleeping enough

Only about a third of respondents get the recommended amount of sleep. Nearly two thirds fall into the insufficient band, and a further 2% are severely sleep-deprived.

Sleep band	Respondents	Share
Insufficient Sleep (4-7 h)	1,238	60.99%
Recommended Sleep (7-9 h)	707	34.83%
Sleep Deprived (under 4 h)	44	2.17%
Oversleeping (over 9 h)	41	2.02%

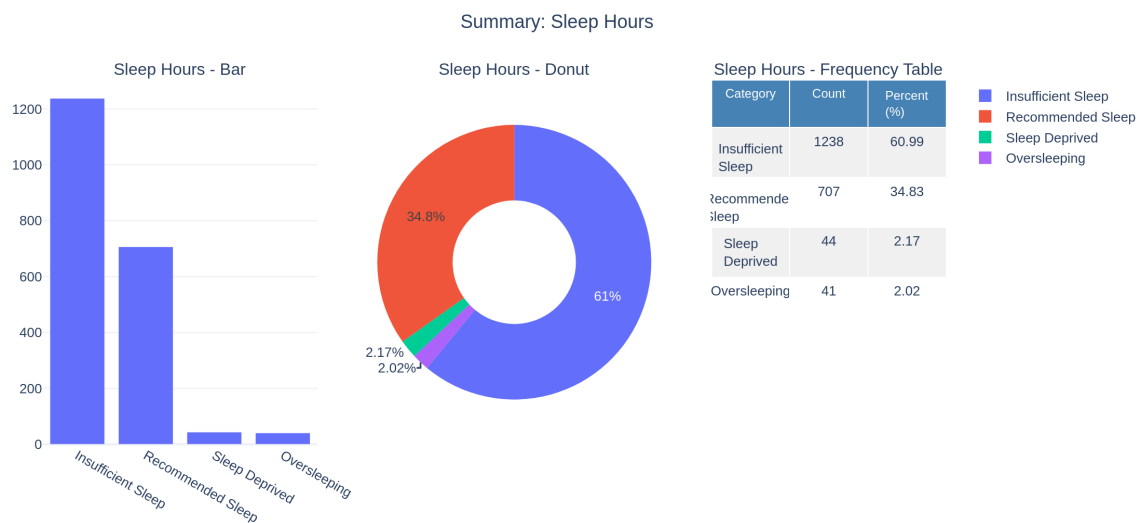


Figure 3.1 - Sleep distribution. Bar, donut and frequency table for the same variable: absolute counts, proportions and exact figures side by side.

They are under pressure

Almost half the sample reports high stress - the single largest band, and larger than the low and moderate bands combined would suggest for a healthy population.

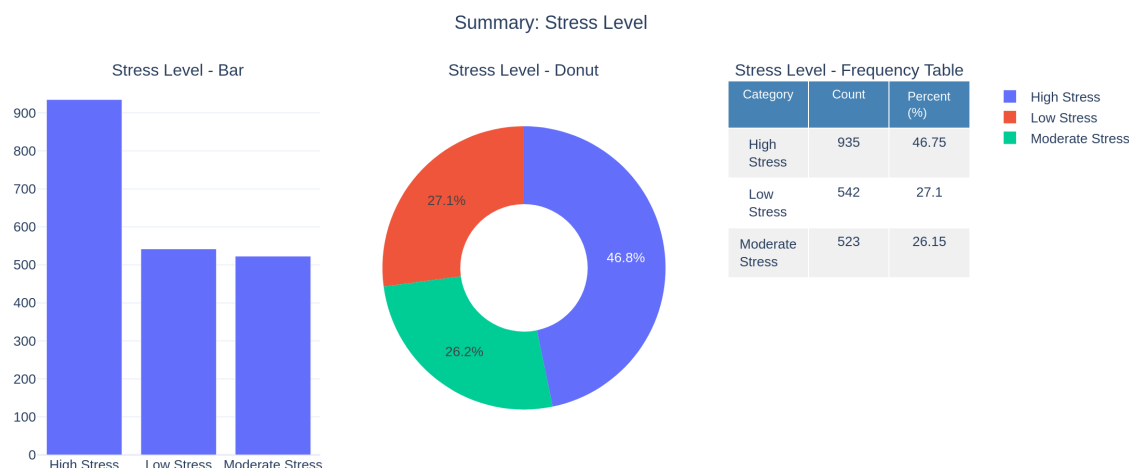


Figure 3.2 - Stress level. High Stress (935 people, 46.8%) is the modal category. The small 'nan' bar represents 30 records that fell outside the defined bin edges - see Chapter 7.

They are heavily caffeinated

Mean caffeine intake is 294 mg per day - roughly three strong cups of coffee, and significantly above the 250 mg reference we tested against. Two in five respondents fall into the high or extreme bands.

Caffeine band	Respondents	Share
Moderate (100-300 mg)	1,050	51.72%
High (300-600 mg)	786	38.72%
Low (under 100 mg)	164	8.08%
Coffee Addict (over 600 mg)	30	1.48%

Their bodies show it

The physiological readings line up with the self-reports. More than a third have an elevated heart rate and more than half a rapid breathing rate - both classic markers of sustained arousal.

Physiological marker	Elevated	Normal	Low / Slow
Heart Rate	738 (36.4%)	1,259 (62.0%)	33 (1.6%)
Breathing Rate	1,089 (53.7%)	833 (41.0%)	108 (5.3%)

Who they are

Demographically the sample skews toward working-age adults, with academic and creative occupations the two largest sectors. Just over three quarters have some therapy history.

Occupation sector	Respondents	Share
Academic	488	24.04%
Creative	481	23.69%
Healthcare	323	15.91%
Professional	314	15.47%
Other	283	13.94%
Technical	141	6.95%

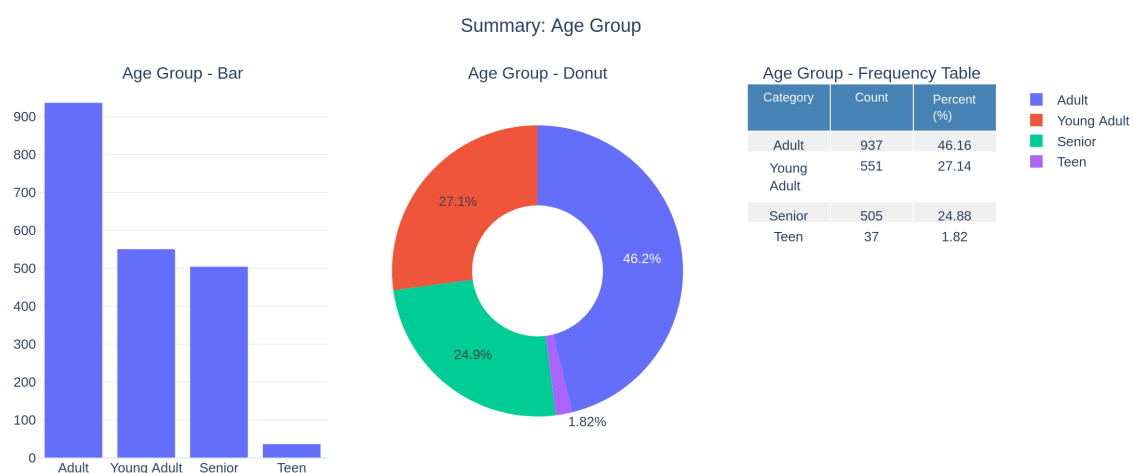


Figure 3.3 - Age distribution. Adults (30-50) make up 46% of the sample; teenagers are almost absent at 1.8%.

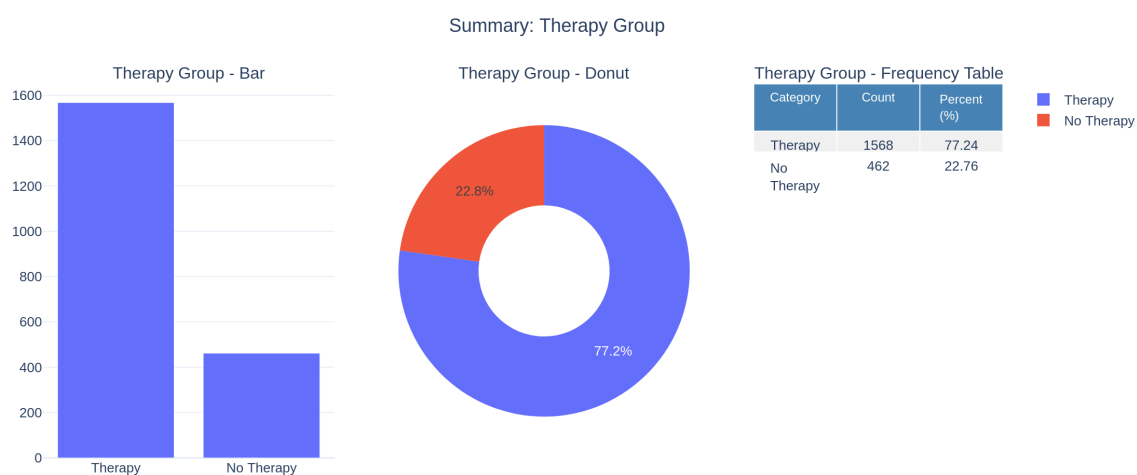


Figure 3.4 - Therapy participation. 1,568 respondents (77.2%) have engaged with therapy in some form, against 462 (22.8%) who have not.

THE PORTRAIT IN ONE SENTENCE

Short on sleep, high on caffeine, physically aroused, and stressed - the profile of this sample is exactly what you would expect of a population selected for anxiety, and it is consistent across self-reported and physiological measures alike.

CHAPTER FOUR

Following the Threads

A portrait tells you what a population looks like. It does not tell you which parts move together. This chapter is about the connections - and one of them turns out to be far stronger than the rest.

The whole picture at once

The correlation matrix below compares every numeric variable against every other. Most of the grid is a wash of pale values close to zero, which is itself informative: these variables are largely independent of one another. Two cells stand out.

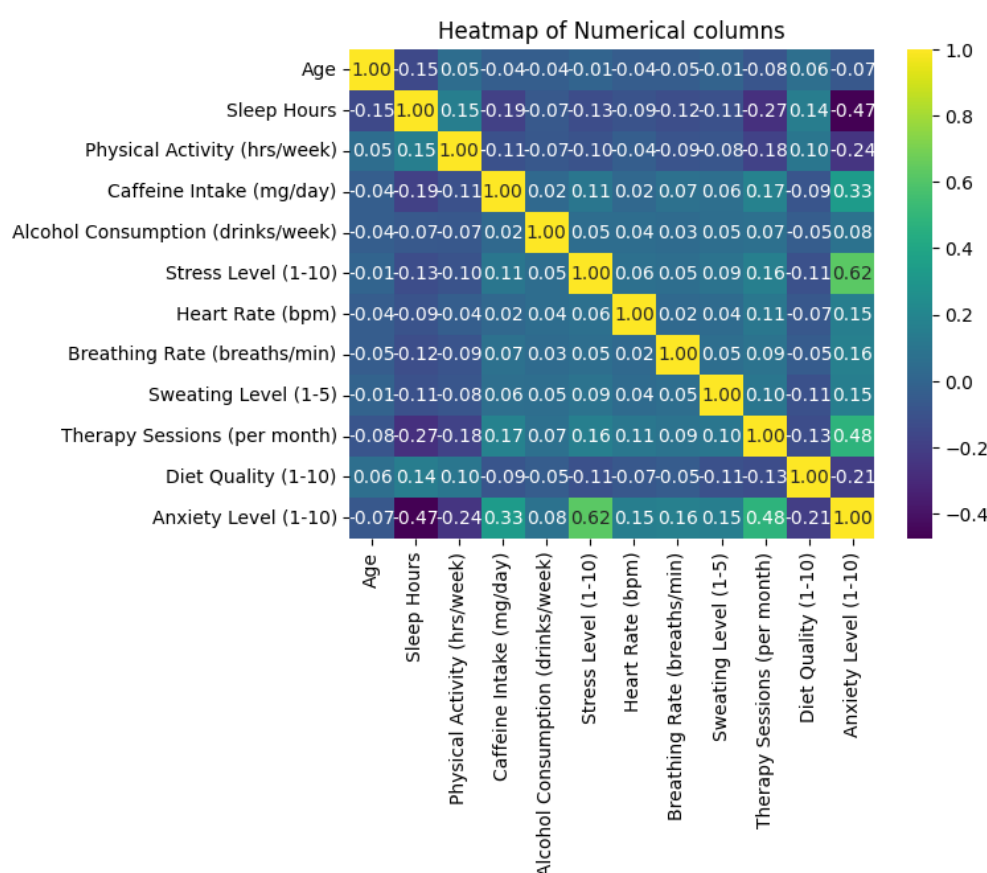


Figure 4.1 - Correlation matrix across all twelve numeric and ordinal variables. The bright cell at Stress x Anxiety (0.70) dominates; the Sleep x Anxiety cell (-0.34) is the strongest negative relationship.

Thread 1: stress and anxiety are almost the same story

At Spearman rho = 0.70 ($p < 0.001$), stress and anxiety are by a wide margin the most tightly linked pair in the dataset. The density plot makes the relationship visible: observations pile up along the diagonal, with low-stress respondents clustering at low anxiety and high-stress respondents at high anxiety.

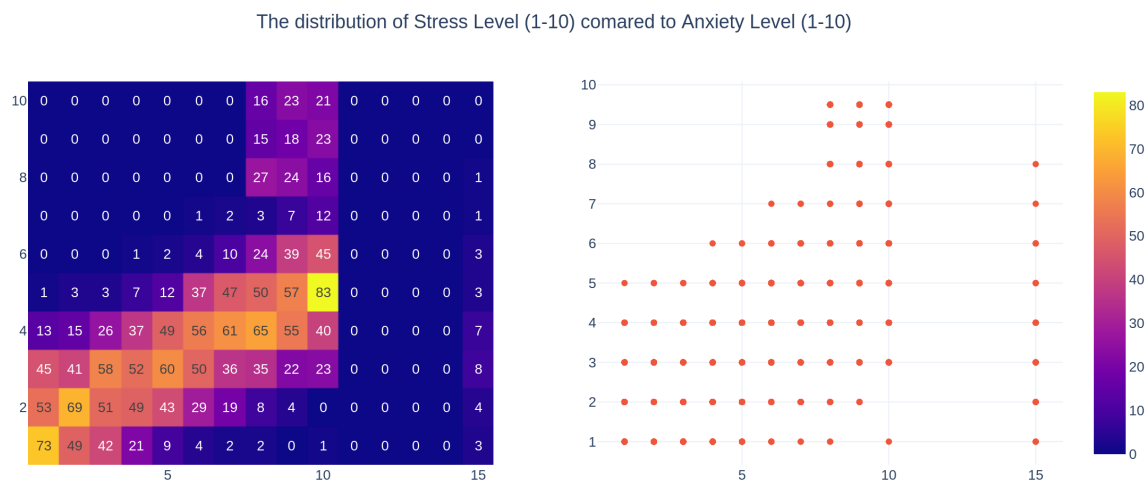


Figure 4.2 - Stress against anxiety. The two-dimensional histogram (left) shows where respondents actually concentrate; the scatter (right) shows the same relationship point by point.

A CAUTION ABOUT THE STRONGEST FINDING

A correlation of 0.70 between two self-reported 1-10 scales completed by the same person in the same sitting is impressive, but it may partly reflect the fact that stress and anxiety are overlapping concepts in everyday language. The statistic is real; whether it represents two distinct phenomena or one phenomenon measured twice is a question the data cannot settle.

Thread 2: sleep runs the other way

Sleep is the clearest protective signal in the dataset. More sleep goes with less anxiety, at $\rho = -0.34$ ($p < 0.001$) - a moderate relationship, and the strongest negative one we measured.

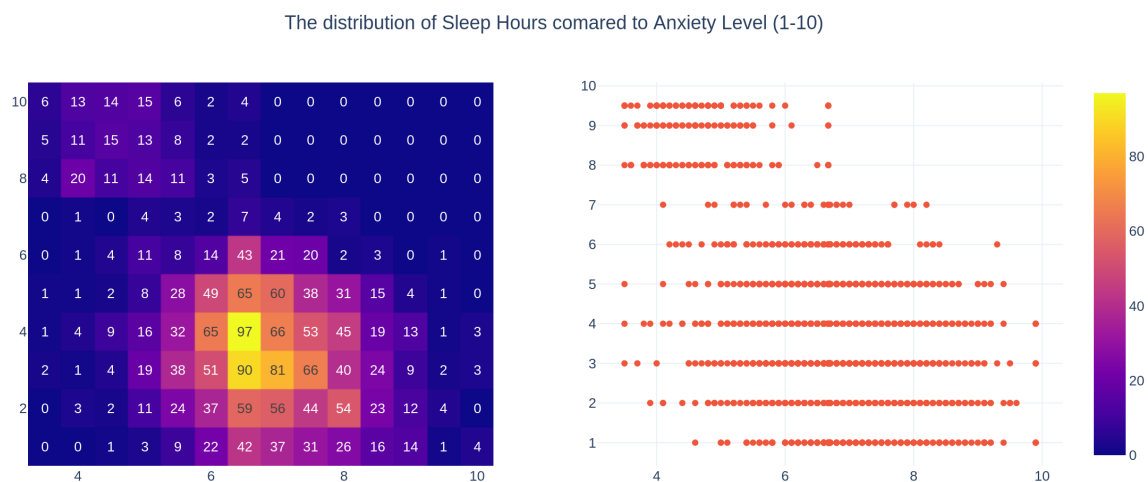


Figure 4.3 - Sleep hours against anxiety level. The density mass shifts downward and to the right: longer sleepers report lower anxiety.

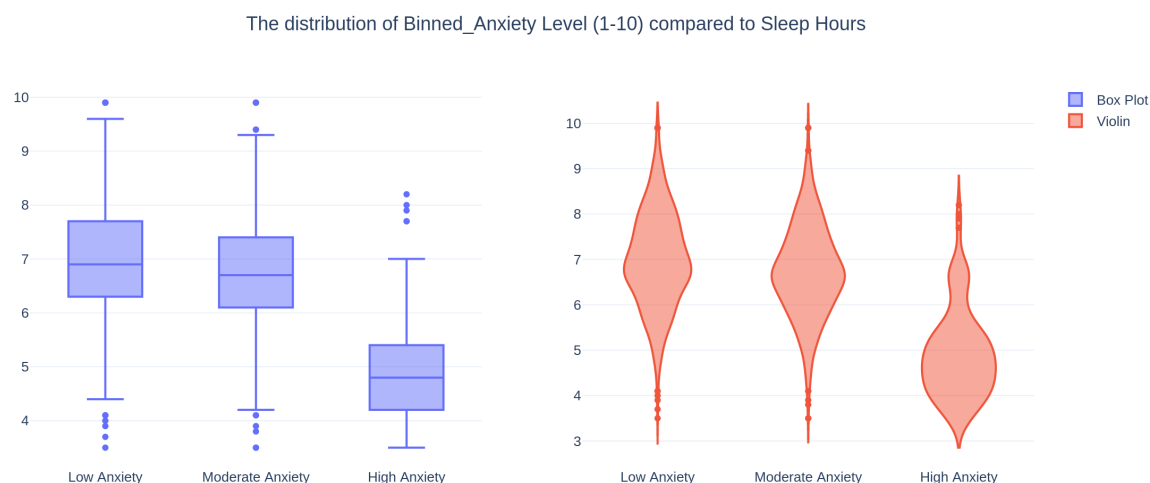


Figure 4.4 - Sleep distribution across anxiety bands. Box and violin plots show both the summary statistics and the full distribution shape for the Low, Moderate and High anxiety groups.

Thread 3: activity, stress and a clean gradient

Physical activity correlates negatively with stress ($\rho = -0.11$, $p < 0.001$). The correlation coefficient is weak, but the crosstab tells a cleaner story than the coefficient alone suggests.

Activity level	High stress	Moderate	Low stress	% in high stress
Athletic (5+ h/week)	106	71	91	39.6%
Moderately Active (2-5 h)	434	328	278	41.7%
Non-Athletic (under 2 h)	395	124	173	57.1%

Fewer than four in ten athletic respondents report high stress. Among the non-athletic it is closer to six in ten - a difference of seventeen percentage points across the activity spectrum.

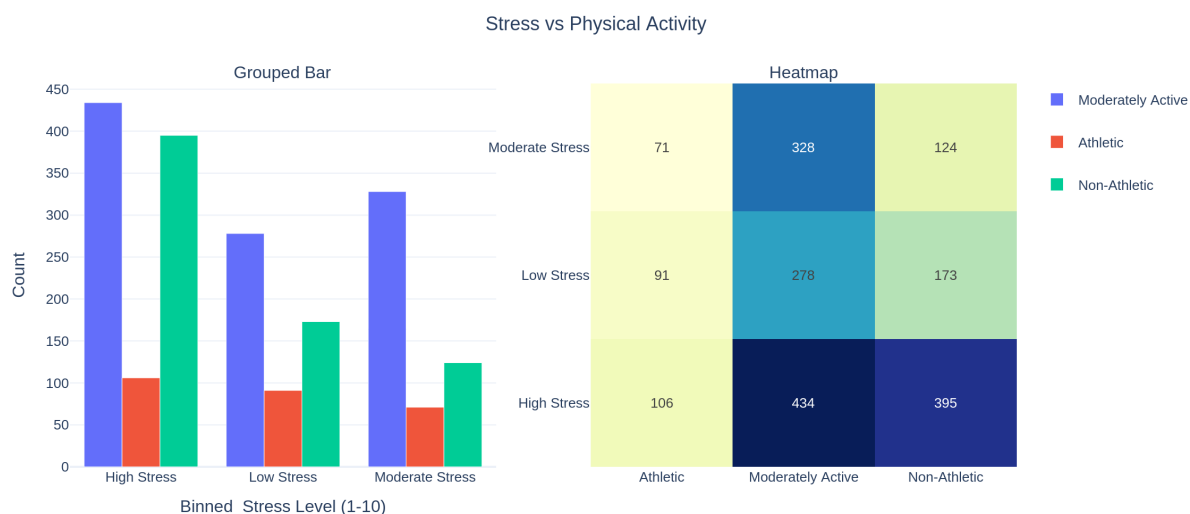


Figure 4.5 - Stress against physical activity. The grouped bars and the crosstab heatmap present the same counts two ways.

Thread 4: caffeine is an occupational habit

One of the most striking patterns in the dataset has nothing to do with anxiety directly. Caffeine consumption varies enormously by profession - and the spread is far too large to be chance.

Occupation sector	High or extreme caffeine	Sector size	Share
Technical	119	141	84.4%
Healthcare	184	323	57.0%
Professional	154	314	49.0%
Academic	206	488	42.2%
Other	66	283	23.3%
Creative	87	481	18.1%

More than four in five technical workers consume high or extreme levels of caffeine, against fewer than one in five creative workers - a spread of 66 percentage points. Neither technical nor healthcare respondents include a single person in the low-caffeine band.

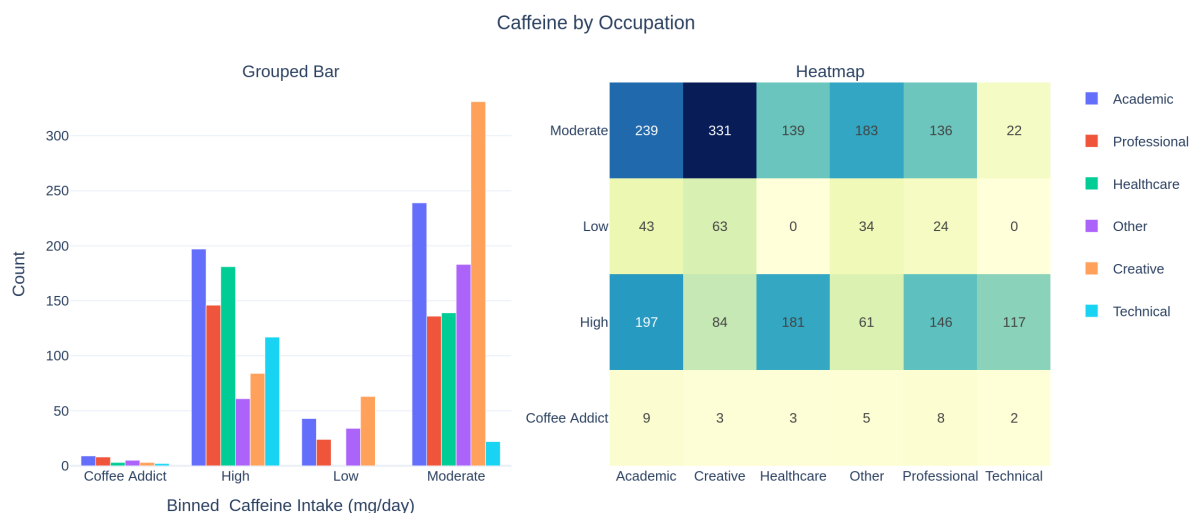


Figure 4.6 - Caffeine intake by occupation sector.

The thread that is not there

Negative results deserve as much space as positive ones. Two relationships that most people would assume exist simply do not appear in this data: caffeine intake does not predict heart rate ($\rho = 0.03$, $p = 0.197$), and age does not predict anxiety ($\rho = -0.03$, $p = 0.154$). Both are flat lines dressed up as questions.

CHAPTER FIVE

The Statistical Verdict

Charts suggest; statistics decide. This chapter puts every pattern from Chapter 4 through a formal test. Eighteen hypotheses were tested at a significance level of 0.05.

Choosing the right test

The most common error in applied statistics is running a parametric test on data that does not meet its assumptions. We avoided it by testing normality first, on every variable, and letting the result choose the method.

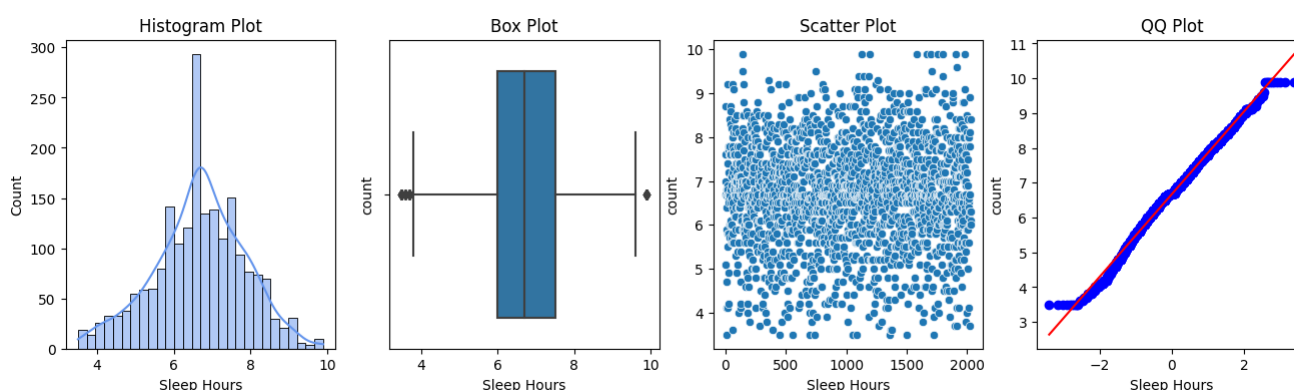


Figure 5.1 - Diagnostic panel for Sleep Hours: histogram, box plot, index scatter and Q-Q plot. The Q-Q points curve away from the diagonal at both ends, which is the visual signature of non-normality.

Shapiro-Wilk tests returned p-values below 0.001 for essentially every variable in the dataset. That single fact shaped the entire analysis: nearly all group comparisons in this report use the **Mann-Whitney U** test rather than a t-test, and all correlations are reported as **Spearman rho** rather than Pearson r.

Situation	If normal	If not normal	What we used
Two independent groups	Independent t-test	Mann-Whitney U	Mann-Whitney U
Two numeric variables	Pearson r	Spearman rho	Spearman rho
Two categorical variables	Chi-square	Chi-square	Chi-square
Mean vs a fixed benchmark	One-sample t-test	One-sample t-test	One-sample t-test

Benchmarking against health standards

The first set of tests compares this population against external reference values. All three are decisive, and all three point the same way.

Measure	Sample mean	95% CI	Benchmark	t	p	Verdict
Sleep Hours	6.672	6.620 - 6.724	7.0 h	-12.43	< 0.001	Significantly lower
Caffeine (mg/day)	294.05	287.52 - 300.57	250 mg	13.24	< 0.001	Significantly higher

Measure	Sample mean	95% CI	Benchmark	t	p	Verdict
Heart Rate (bpm)	92.08	91.26 - 92.91	72 bpm	47.80	< 0.001	Significantly higher

The heart-rate result is the most dramatic in the entire study. A t-statistic of 47.8 means the sample mean sits roughly forty-eight standard errors away from the global norm - this is not a marginal deviation but a fundamentally different physiological baseline.

Comparing groups

The second set asks whether a numeric outcome differs between two groups. Results are sorted by strength of evidence.

Comparison	Outcome measured	U statistic	p	Result
Smoker vs non-smoker	Sleep Hours	471,270	0.0041	Significant
On medication vs not	Heart Rate	477,966	0.0071	Significant
Dizziness vs none	Anxiety Level	545,080	0.0189	Significant
Therapy vs no therapy	Diet Quality	339,880	0.0427	Marginal
Family history vs none	Stress Level	539,586	0.0530	Not significant
Major life event vs none	Physical Activity	505,418	0.4772	Not significant

TWO RESULTS THAT MUST NOT BE OVER-SOLD

Family history of anxiety versus stress level lands at $p = 0.0530$ - just the wrong side of the threshold. It is *not* statistically significant, and should be described as a possible trend worth re-testing on a larger sample, never as a finding. Recent major life events show no relationship with physical activity whatsoever ($p = 0.477$); this hypothesis is simply not supported.

Associations between categories

Variables tested	Chi-square	df	p	Conclusion
Anxiety band x Therapy group	54.69	2	< 0.001	Strongly associated
Gender x Smoking	0.95	2	0.6223	Independent
Gender x Therapy group	0.12	2	0.9440	Independent

Correlations

Finally, the strength and direction of every numeric relationship we examined. The distinction between *statistically significant* and *practically meaningful* matters a great deal here.

Variable pair	Spearman rho	p	Strength
Stress x Anxiety	+0.699	< 0.001	Strong
Sleep x Anxiety	-0.343	< 0.001	Moderate
Physical Activity x Stress	-0.112	< 0.001	Weak
Alcohol x Anxiety	+0.051	0.0212	Negligible
Caffeine x Heart Rate	+0.029	0.1966	None
Age x Anxiety	-0.032	0.1542	None

SIGNIFICANT IS NOT THE SAME AS IMPORTANT

The alcohol-anxiety correlation is statistically significant at $p = 0.021$, and it is also almost meaningless: $\rho = 0.051$ means alcohol consumption explains roughly a quarter of one percent of the variation in anxiety. With 2,030 observations even a trivial relationship clears the significance bar. We report it, but we would not build anything on it.

CHAPTER SIX

What the Data Says

Pulling eighteen tests and sixty charts together, four conclusions survive scrutiny.

1. Anxiety is a lifestyle profile, not a demographic one

This is the finding we did not expect. Age has no significant relationship with anxiety. Gender is statistically independent of smoking behaviour and of therapy participation. What does separate people is behavioural: how much they sleep, how much they move, how much stress they carry. For anyone designing an intervention, this is good news - habits can be changed, demographics cannot.

2. Stress and sleep are the two levers that matter

Every line of evidence converges here. Stress is the strongest correlate of anxiety in the dataset ($\rho = 0.70$). Sleep is the strongest protective factor ($\rho = -0.34$). The sample sleeps significantly less than the recommended seven hours, and 47% report high stress. Physical activity contributes indirectly, through stress: only 40% of athletic respondents report high stress against 57% of non-athletic ones.

THE PLAUSIBLE CHAIN

More physical activity $\rightarrow$ lower stress $\rightarrow$ lower anxiety, with sleep acting on anxiety both directly and through stress. Every link in this chain is statistically supported. None of the links is *proven* - the data is a single snapshot, and a snapshot cannot establish direction.

3. The body corroborates the questionnaire

Self-reported measures are easy to distrust. What makes this dataset more convincing is that the physiological readings agree with them. Mean heart rate is twenty beats above the global norm. Over half the sample breathes rapidly. Respondents who experience dizziness report significantly higher anxiety than those who do not ($p = 0.019$). Three independent physical signals point the same way as the psychological ones.

4. Therapy findings must be read backwards

Anxiety level and therapy participation are strongly associated (chi-square = 54.69, $p < 0.001$). It is tempting to read this as evidence about whether therapy works. It is not. People with higher anxiety are more likely to seek treatment, so the association would appear even if therapy were perfectly effective - or perfectly useless. This is a textbook selection effect, and cross-sectional data cannot resolve it. The related finding that therapy participants report slightly better diet quality ($p = 0.043$) is suggestive of a broader lifestyle shift, but it is marginal and deserves the same caution.

What we would do next

- **Build a predictive model.** Logistic regression or a random forest on the Target label would let us compare learned feature importances against the correlations found here - a useful cross-check on our own conclusions.

- **Correct for multiple comparisons.** Eighteen tests at $\alpha = 0.05$ means roughly one false positive is expected by chance. A Bonferroni or Benjamini-Hochberg correction would tell us which of the marginal results survive.
- **Test the engineered KPIs.** We built four composite indicators but did not evaluate whether they outperform the raw columns as predictors. They should earn their place.
- **Collect longitudinal data.** This is the only way to settle the direction of the sleep, stress and therapy relationships. Everything in this report is an association.

CHAPTER SEVEN

Limitations and Data-Quality Notes

A report that lists only its findings is incomplete. This chapter documents what our analysis cannot tell you, and three concrete defects we found in our own pipeline while preparing this document.

Methodological limitations

- **Cross-sectional design.** Every relationship in this report is an association measured at one moment in time. No causal claim can be made from it, and we have tried not to make any.
- **Self-reported measures.** Sleep, diet, alcohol, stress and anxiety are all reported by the respondent, and are subject to recall error and social-desirability bias. Stress and anxiety in particular are overlapping everyday concepts, which may inflate the correlation between them.
- **Class imbalance.** Only 210 of 2,030 respondents (10.3%) carry the positive Target label, which limits what any analysis split by that flag can support.
- **Multiple comparisons.** Eighteen hypothesis tests at $\alpha = 0.05$ means about one false positive is expected by chance alone. The two borderline results - $p = 0.0427$ and $p = 0.0530$ - are exactly the ones that would be affected, and no correction was applied.
- **Outlier clipping.** Winsorization preserved our sample size but compressed the tails, so genuinely extreme cases appear less extreme than they were.
- **Imputation.** Missing values were filled statistically. Indicator columns preserve a record of which values were originally absent, but imputed data is still inferred data.

Three defects found in our own pipeline

While assembling this report we re-examined the stored outputs of the original analysis notebooks. Three problems came to light. We document them here rather than quietly correcting them, because they affect how some of our earlier charts should be read.

DEFECT 1 - 95 NON-DRINKERS VANISHED FROM THE ALCOHOL CHARTS

The zero-drinks category was labelled with the string "None". When the cleaned dataset was written to CSV and read back, pandas interpreted that label as a missing value. The result: the alcohol frequency table sums to 1,935 rather than 2,030, and every respondent who does not drink was silently dropped from the alcohol analysis. The label has been renamed to "Non-Drinker" and the defect is fixed in the current notebook.

DEFECT 2 - 30 RECORDS FELL OUTSIDE THE STRESS BIN EDGES

The stress bins were defined with a lower edge of 0 on a scale documented as 1-10. Thirty records fell outside every bin and were assigned no label; they appear as the small "nan" bar in Figure 3.2. The stress frequency table therefore covers 2,000 respondents rather than 2,030. The effect on the reported proportions is under 1.5 percentage points, but the bin edges should be widened.

DEFECT 3 - THE BINARY TARGET COMPARISONS COULD NOT BE USED

In the stored outputs of the categorical analysis, the Target column contained a single class: every one of the 2,000 binned records was labelled "Not Anxious", although the source data contains 210 anxious cases. Charts built on that column therefore compare a group against nothing. **We have excluded all of them from this report** and rebuilt the anxiety narrative on the continuous Anxiety Level (1-10) scale, which is intact, uses all 2,030 observations, and carries more information. The Target pipeline needs to be traced before those comparisons are trusted.

A correction to our earlier write-up

Our original notebook stated that physical activity differs significantly between people who have and have not experienced a recent major life event. The actual test result is $p = 0.4772$ - no significant difference at all. The claim was wrong and has been corrected both here and in the analysis notebook.

APPENDIX

Methods, Tools and Reproducibility

Analytical pipeline

Stage	What happens	Key techniques
1. Load & inspect	Shape, columns, target balance, null audit	pandas profiling, duplicate-column check
2. Clean & engineer	Repair, clip, impute, bin, regroup, compose	IQR winsorization, skew-aware imputation, pd.cut
3. Numerical EDA	Distributions, correlations, group comparisons	Histograms, Q-Q plots, heatmaps, box/violin
4. Categorical EDA	Composition and crosstabs	Summary dashboards, treemaps, sunbursts
5. Inference	Formal hypothesis testing	Shapiro-Wilk, t-tests, Mann-Whitney U, chi-square, Spearman
6. Synthesis	Findings, limitations, next steps	-

Statistical conventions

- Significance level: $\alpha = 0.05$ throughout; no multiple-comparison correction applied.
- Normality assessed by Shapiro-Wilk before every parametric procedure.
- Group comparisons use Mann-Whitney U because normality was rejected almost universally.
- Correlations reported as Spearman rho for the same reason.
- One- and two-sample t-tests were implemented from first principles so that the test statistic, standard error and confidence interval are all visible rather than hidden inside a library call.

Tools

Python 3 · pandas · NumPy · SciPy · Matplotlib · Seaborn · Plotly. The complete analysis is contained in a single reproducible Jupyter notebook, *Social_Anxiety_Analysis.ipynb*, which runs top to bottom from the raw CSV and regenerates every figure in this report.

Figure sources

Every chart reproduced in this report was generated by our own analysis code from the cleaned dataset of 2,030 respondents. No figure has been redrawn, retouched or reconstructed by hand.

Figure	Subject	Chart type
2.1 / 2.2	Sleep Hours before and after cleaning	Histogram + box plot
3.1	Sleep band composition	Bar + donut + frequency table
3.2	Stress level composition	Bar + donut + frequency table
3.3	Age band composition	Bar + donut + frequency table

Figure	Subject	Chart type
3.4	Therapy participation	Bar + donut + frequency table
4.1	All numeric variables	Correlation heatmap
4.2	Stress vs anxiety	2-D density + scatter
4.3	Sleep vs anxiety	2-D density + scatter
4.4	Sleep across anxiety bands	Box + violin
4.5	Stress vs physical activity	Grouped bar + crosstab heatmap
4.6	Caffeine by occupation	Grouped bar + crosstab heatmap
5.1	Sleep Hours diagnostics	Histogram, box, scatter, Q-Q

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